

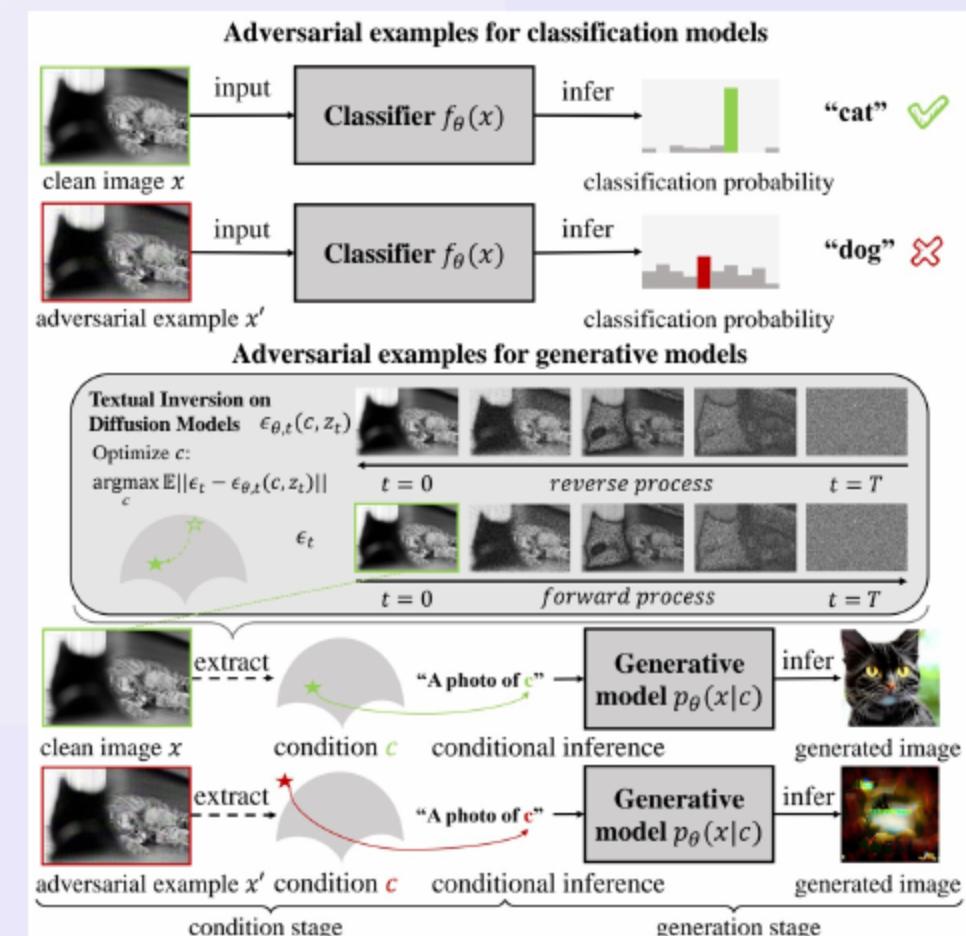
Adversarial Example Does Good: Preventing Painting Imitation from Diffusion Models via Adversarial Examples

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01 Title & Authors

- Comprehensive author team from multiple institutions
- Collaborative work spanning systems and machine learning
- Research focuses on safeguarding artistic originality

02 Motivation: Protecting Art



- Goal: imperceptible perturbations hinder imitation
- Framework + AdvDM using Monte Carlo paths
- Strong protection across datasets and tools

03 Why Classifier Attacks Fail

- Classifier losses differ from diffusion objectives
- Diffusion relies on indirect latent conditioning
- Prior attacks transfer poorly to diffusion

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Background & Framing

- Generative models learn data distributions
- Conditional $p_{\theta}(x|c)$ for styles/text
- Latent diffusion is efficient and faithful

05

Adversarial Framing

- Perturb artworks without perceptual change
- Disrupt condition extraction during denoising
- Aim for consistent disruption across paths

06

Evaluation Setup

DATASET METRIC	LSUN-CAT			LSUN-SHEEP			LSUN-AIRPLANE		
	FID $\uparrow$	<i>prec.</i> $\downarrow$	<i>recall.</i>	FID $\uparrow$	<i>prec.</i> $\downarrow$	<i>recall.</i>	FID $\uparrow$	<i>prec.</i> $\downarrow$	<i>recall.</i>
NO ATTACK	34.94	0.5643	0.1531	32.81	0.6378	0.1228	39.22	0.5016	0.2765
ADVDM	127.04	0.1708	0.061	203.5	0.0058	0.378	169.67	0.0263	0.3235

- Scenarios: textual inversion, style, img2img
- Datasets: LSUN, WikiArt, curated photos
- Metrics: FID, Precision, Recall; Models: SD/LDM

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AdvDM: Concept Only

- Concept: leverage diverse denoising paths for a single perturbation
- Outcome: consistently weakens conditioning across prompts/tools

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Textual Inversion Results

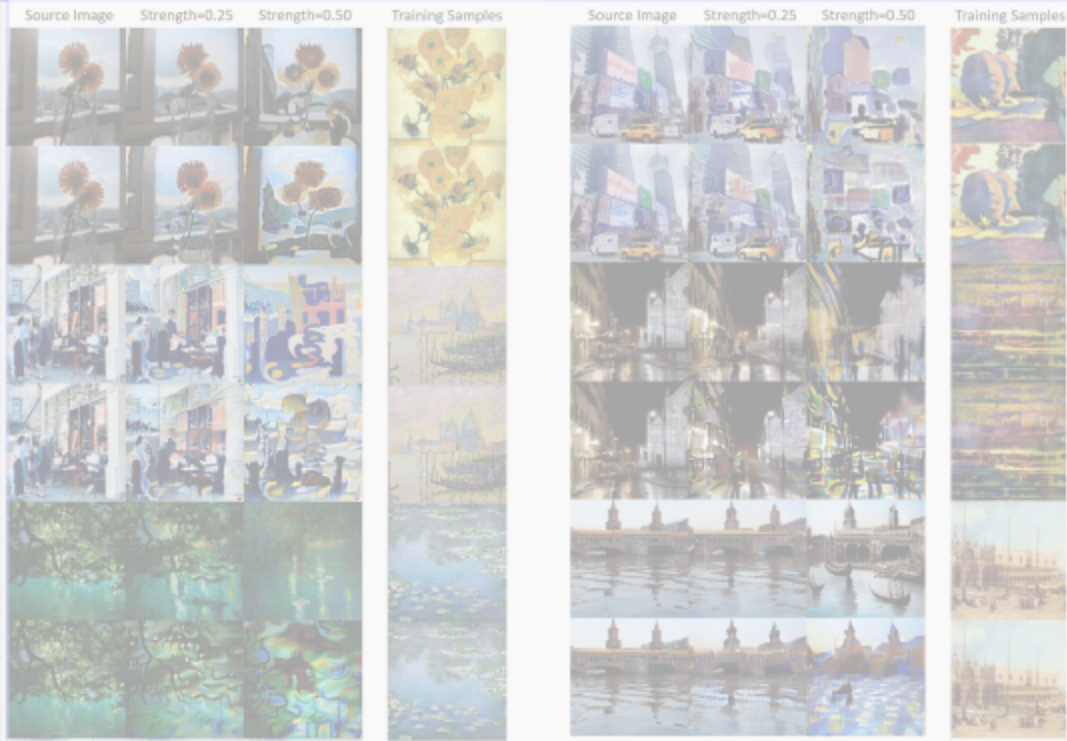
- Quality drops on LSUN Sheep
- Object capture becomes generic
- Generations revert to generic animals

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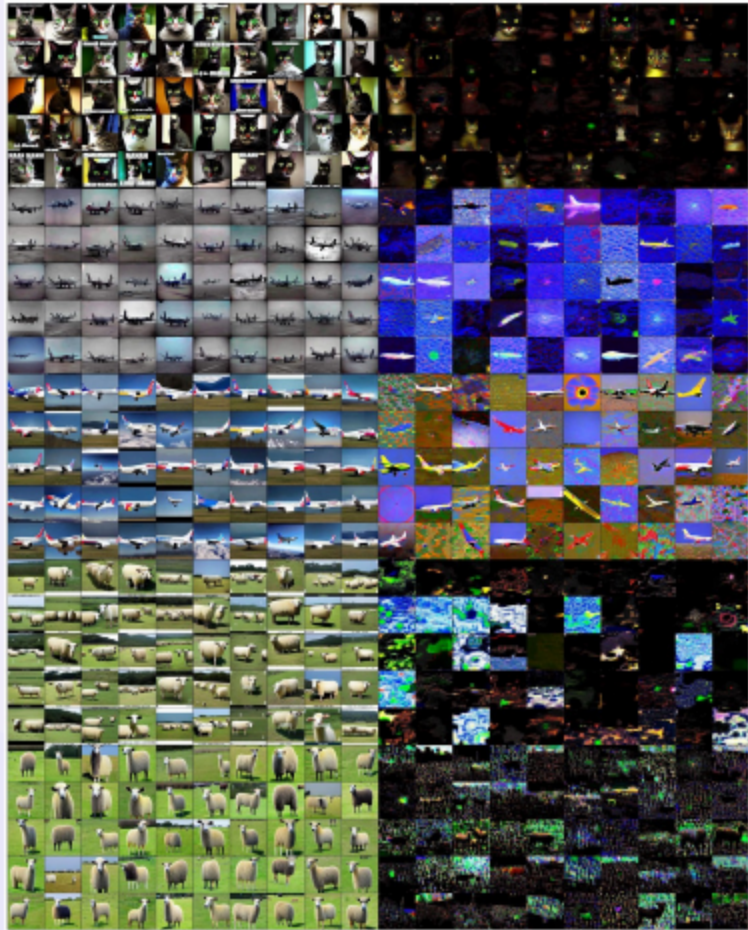
Style Transfer Results

- Protected training loses brushwork and texture
- Styles drift to generic palettes



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Image-to-Image Results



- Structure and alignment frequently fail
- Colors bleed; textures wash out

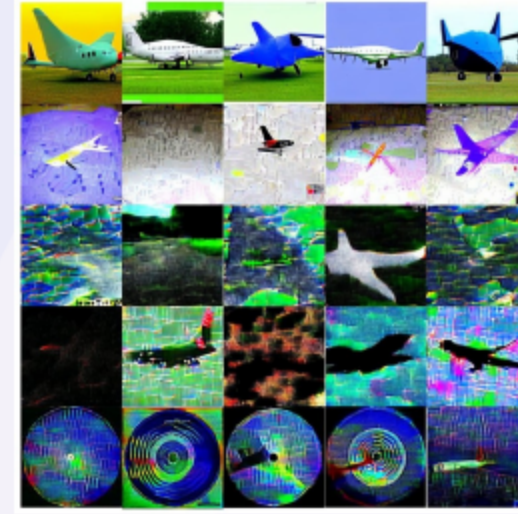
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Related Work & Defenses

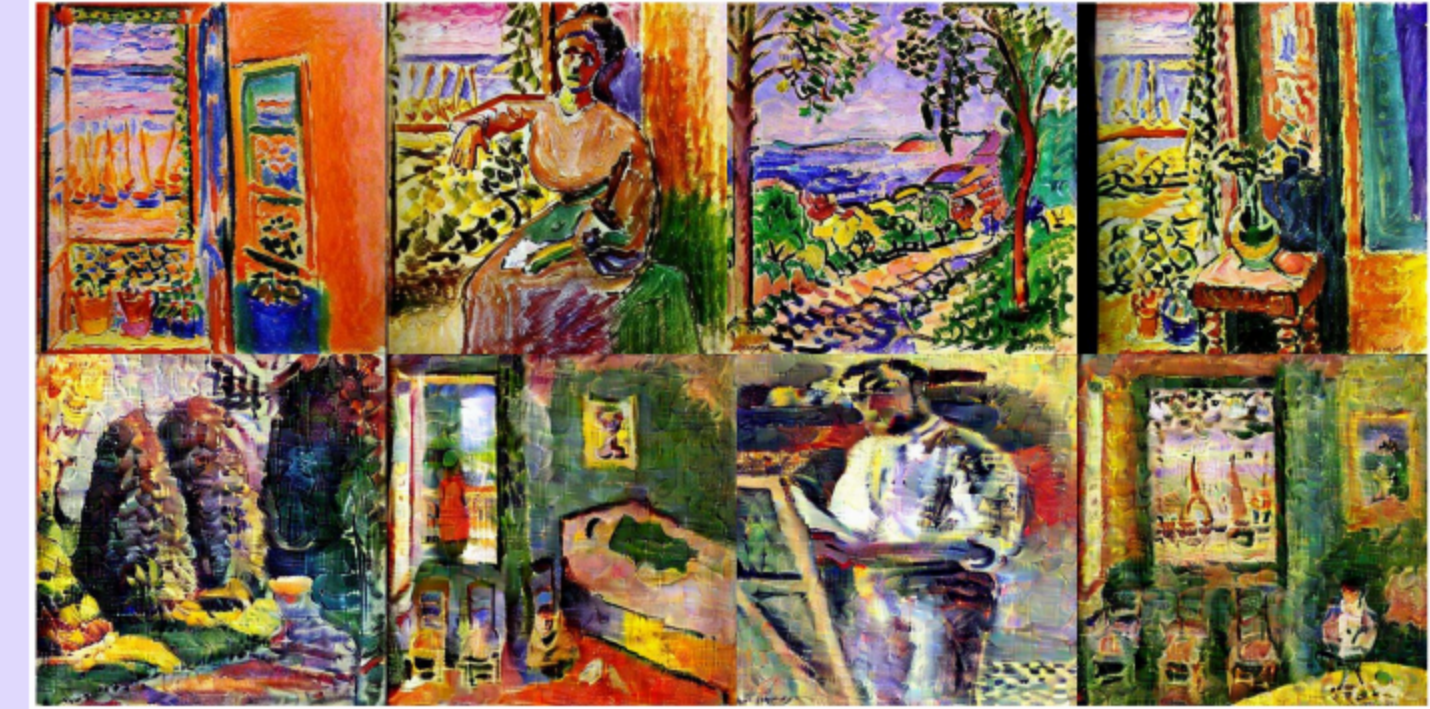
- Prior adversarial work targets classifiers and does not fit diffusion conditioning
- Preprocessing and denoising defenses offer **limited recovery** in practice
- Qualitative assessments indicate persistent protection under defenses



- More Monte Carlo steps strengthen protection with higher compute cost
- Larger budgets improve disruption but risk **visibility** artifacts
- Simple preprocessing helps partially but protection gaps remain



SAMPLING STEPS	FID↑	<i>prec.</i> ↓	INFERENCE TIME
10	122.9	0.05	1.803
40	186.05	0.037	6.342
1000	211.88	0.011	166.6



- Adversarial protection offers **practical** support for copyright preservation
- Residual artifacts and partial recoveries present **open challenges**
- Future directions include stronger defenses and perceptual safeguards

Thank You

Questions & Discussion